PROJECT REPORT

FOR

Foundations of Artificial Intelligence

INTELLIGENT ROUTE PLANNER WITH MULTI-MODAL TRANSPORT

A PROJECT REPORT

Submitted by

2116240701439 – Roshan Bright R

2116240701629 – Immanuel SA Spurgeon



October, 2026

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ABSTRACT

Urban transportation systems involve multiple modes of travel such as walking, cycling, buses, and metro services. Selecting an efficient route becomes difficult when a journey requires more than one mode of transportation. Conventional route-planning systems generally focus on finding the shortest or fastest route and may not adequately consider factors such as traffic conditions, public transportation schedules, waiting time, transfer time, travel cost, walking distance, and user preferences.

The system represents the transportation network as a multi-modal graph in which different modes such as walking, cycling, bus, and metro are connected through suitable transfer points. It applies **Dijkstra's Algorithm** to determine minimum-cost or shortest paths, **A* Algorithm** to improve route searching using heuristic information, and **Anytime Weighted A*** to obtain a good-quality solution quickly and progressively improve the solution when additional computation time is available. These algorithms enable the system to handle different routing requirements while reducing unnecessary exploration of the transportation network.

An important feature of the proposed system is its consideration of **time-dependent travel conditions**. Travel time and route cost may vary depending on the departure time, traffic conditions, public transportation schedules, waiting periods, and transfers between transportation modes. By incorporating these factors into route evaluation, the system can provide routes that are more practical for real-world journeys. The system also uses a **multi-objective optimization approach** to evaluate routes based on parameters such as total travel time, transportation cost, walking distance, number of transfers, and comfort.

The proposed planner allows users to specify their source, destination, departure time, and preferred travel criteria such as speed, cost, or comfort. Based on these inputs, multiple suitable routes can be generated and ranked as **fastest, cheapest, most comfortable, or balanced routes**. An interactive map can then be used to visualize the selected route and display the sequence of transportation modes, estimated travel time, cost, and other relevant information

1.INTRODUCTION

Transportation plays an important role in the daily activities of people living in modern cities. As urban populations continue to grow, people increasingly depend on different modes of transportation such as **walking, cycling, buses, and metro services** to reach their destinations. In many real-world journeys, a person may need to use more than one mode of transportation. For example, a passenger may walk from their home to a bus stop, travel by bus, transfer to a metro, and finally walk to the destination. Selecting the most suitable combination of transportation modes for such journeys can be difficult because several factors need to be considered simultaneously.

Traditional route-planning systems mainly concentrate on finding the **shortest or fastest path** between a source and destination. Although this approach can be useful for simple journeys, it may not provide the most practical route in a multi-modal transportation environment. Factors such as **traffic conditions, public transportation schedules, waiting time, transfer time, travel cost, walking distance, and user comfort** can significantly affect the overall quality of a journey. A route that is shortest in distance may not necessarily be the fastest or most convenient option for a particular user.

The **Intelligent Route Planner with Multi-Modal Transport** is proposed to address these challenges by providing an integrated route-planning approach that considers multiple transportation modes. The system models the transportation network as a multi-modal graph containing walking paths, cycling routes, road networks, bus routes, metro routes, and transfer points. This allows the planner to identify different combinations of transportation modes between the source and destination.

The proposed system uses several pathfinding algorithms to improve route selection. **Dijkstra's Algorithm** is used to determine minimum-cost or shortest paths in the transportation network. **A* Algorithm** improves the search process by using heuristic information to guide the search toward the destination. **Anytime Weighted A*** is used to obtain a good solution quickly and progressively improve the solution when more computation time is available. The combination of these algorithms provides flexibility in handling different route-planning requirements.

2.LITERATURE REVIEW

2.1 Dijkstra's Algorithm

E. W. Dijkstra introduced the shortest-path algorithm in 1959. It determines the minimum-cost path between nodes in a weighted graph and can be used as a baseline for minimum-distance or minimum-cost route calculation.

2.2 A* Algorithm

The theoretical foundation of heuristic search was established by Hart, Nilsson, and Raphael in 1968. A* improves pathfinding by using a heuristic estimate of the remaining distance to the destination, allowing efficient searching in geographical networks.

2.3 Time-Dependent Shortest Paths

Research on time-dependent shortest-path algorithms shows that travel times vary because of traffic congestion and changing transportation conditions. Consequently, static shortest-path methods may be insufficient for accurate multimodal journey planning.

2.4 Multimodal Transportation Planning

Multimodal trip-planning research considers journeys that combine different transportation networks, including walking, road transport, and public transportation. Such systems need to consider transfers and interactions between transportation modes.

2.5 Time-Dependent Intermodal A*

A closely related research direction is time-dependent intermodal A* routing. The selected reference paper, "Time-Dependent Intermodal A* Algorithm: Methodology and Implementation on a Large-Scale Network," studies intermodal routing over transit, walking, and vehicle links while considering realistic time-dependent constraints. This provides academic support for the time-dependent multimodal aspect of the proposed project.

3. EXISTING SYSTEM & PROPOSED SYSTEM

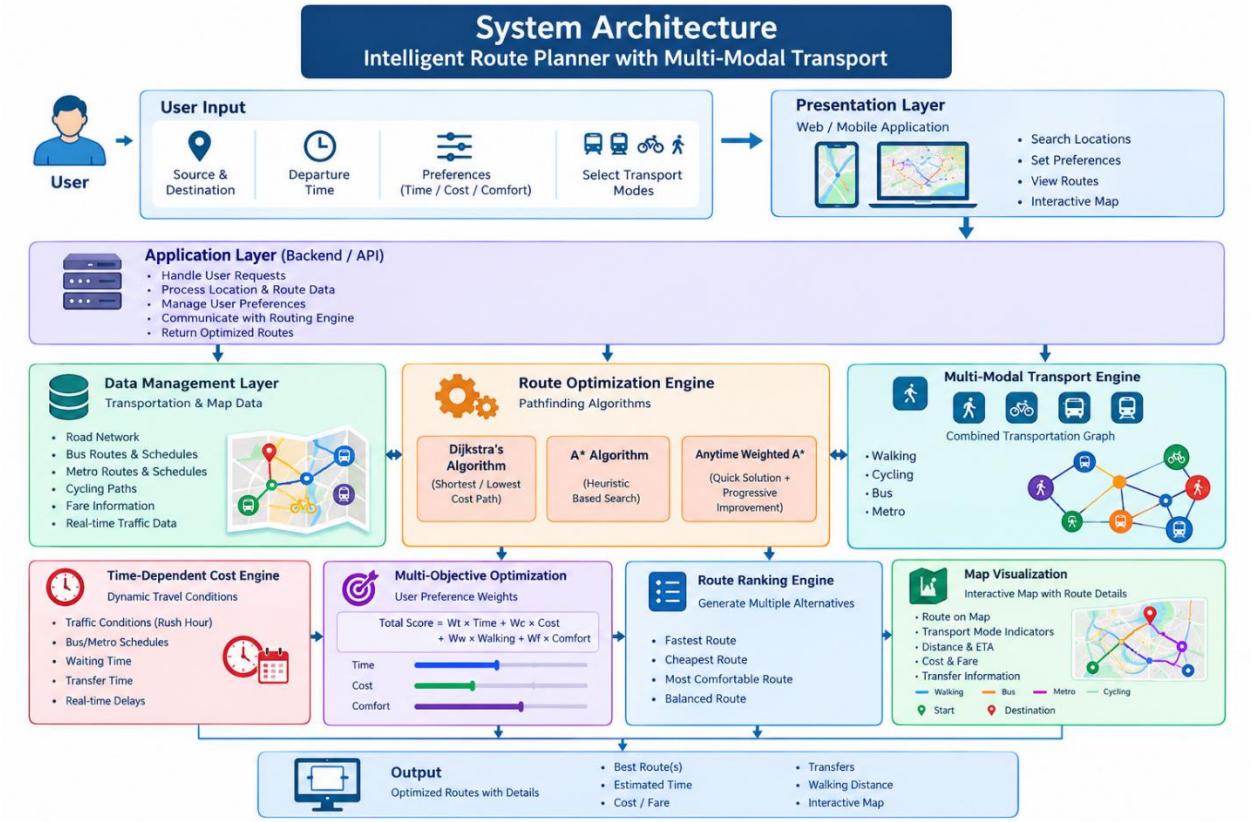
Existing route-planning applications primarily focus on finding a shortest or fastest route between a source and destination. Some systems support multimodal journeys, but the level of integration between walking, cycling, bus, and metro can vary.

- Primary optimization of distance or travel time.
- Limited integration of multiple transportation modes.
- Limited consideration of real-time or time-dependent conditions.
- Incomplete consideration of bus and metro schedules, waiting times, and delays.
- Limited optimization based on cost, comfort, and user preferences.

The proposed system integrates multiple transportation modes and intelligent pathfinding algorithms into a single route-planning framework. The supported modes are walking, cycling, bus, and metro. Dijkstra's Algorithm provides shortest or lowest-cost path calculation, A* provides heuristic-based pathfinding, and Anytime Weighted A* is used for rapid solution generation and progressive improvement.

The system also incorporates departure time, traffic conditions, waiting time, public-transport schedules, and transfer time. Multi-objective optimization evaluates travel time, cost, walking distance, transfers, and comfort. Route alternatives are ranked as fastest, cheapest, comfortable, and balanced according to user preferences.

3.1ARCHITECTURE DIAGRAM



4.MODULES DESCRIPTION

4.1 MODULE 1 – Multi-Modal Transportation Data Management

The Multi-Modal Transportation Data Management module is responsible for collecting, organizing, and managing the transportation information required for route planning. It represents the transportation network as a graph consisting of nodes and edges. Nodes represent locations such as road intersections, bus stops, metro stations, and cycling points, while edges represent connections between these locations.

The module supports multiple transportation modes, including walking, cycling, bus, and metro. Each connection contains information such as distance, estimated travel time, transportation mode, fare, and schedule details. The module also manages departure times, waiting times, and transfer information to support time-dependent route calculations.

The collected data is provided to the route optimization engine, which uses it to generate suitable travel routes. This module serves as the foundation for efficient and accurate multi-modal journey planning.

Key functionalities:

- Collect and manage map and transportation data.
- Represent the transportation network as a graph.
- Store road, bus, metro, and cycling connections.
- Manage distance, travel time, fares, and schedules.
- Provide transportation data to the route optimization engine.

4.2 MODULE 2 – Intelligent Route Optimization and Pathfinding

The Intelligent Route Optimization and Pathfinding module is the core component of the proposed system. It identifies suitable routes between the source and destination using Dijkstra's Algorithm, A* Algorithm, and Anytime Weighted A*. These algorithms evaluate the transportation network and determine routes according to the selected optimization criteria.

Dijkstra's Algorithm is used to calculate minimum-cost paths in a weighted graph. The A* Algorithm uses a heuristic function to guide the search toward the destination efficiently. Anytime Weighted A* initially produces a route quickly and can improve the solution as additional computation time becomes available.

The module also considers time-dependent travel costs, such as rush-hour traffic, public transportation schedules, waiting time, and transfer duration. Multi-objective optimization is applied to balance travel time, transportation cost, walking distance, and comfort according to user preferences.

The final results include alternative routes such as the fastest, cheapest, most comfortable, and balanced routes. These results are passed to the map visualization module for display.

Key functionalities:

- Implement Dijkstra's, A*, and Anytime Weighted A* algorithms.
- Generate routes combining walking, cycling, bus, and metro.
- Calculate route costs based on travel time and distance.
- Consider schedules, waiting times, and traffic conditions.
- Optimize routes based on time, cost, and comfort.
- Rank alternative routes and return the most suitable options.

5. IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The Intelligent Route Planner with Multi-Modal Transport is designed as a web-based application that identifies efficient routes by combining multiple transportation modes. The system integrates graph-based pathfinding algorithms with transportation data and an interactive map interface.

The application can be developed using React for the frontend, Node.js and Express.js for backend services, and Leaflet with OpenStreetMap for map visualization. A database such as MySQL or PostgreSQL can store locations, transportation connections, schedules, and fare information.

The transportation network is represented as a weighted graph, where nodes correspond to locations and edges represent connections between them. Each edge contains relevant attributes such as distance, travel time, transportation mode, fare, and waiting time.

The system implements Dijkstra's Algorithm, A* Algorithm, and Anytime Weighted A* to calculate and compare routes. Time-dependent costs are incorporated to account for traffic conditions, departure times, and public transportation schedules. Multi-objective optimization is used to rank routes according to travel time, cost, walking distance, and comfort.

The system can be evaluated using sample transportation data or available map and public-transport datasets. Different source-destination pairs are tested to compare route quality, execution time, total travel cost, and the number of transfers.

Experimental setup includes:

- Frontend: React.js
- Backend: Node.js and Express.js
- Map visualization: Leaflet and OpenStreetMap
- Database: MySQL or PostgreSQL
- Algorithms: Dijkstra's Algorithm, A*, and Anytime Weighted A*

- Input: Source, destination, departure time, and user preferences
- Evaluation metrics: Travel time, distance, fare, execution time, and route quality

5.2 RESULTS

The proposed system is expected to generate alternative routes by combining walking, cycling, bus, and metro transportation. The user can compare these alternatives and select a route according to personal preferences.

Dijkstra's Algorithm provides a baseline for finding minimum-cost paths, while A* uses heuristic information to guide the search toward the destination. Anytime Weighted A* aims to provide an initial route quickly and improve it as additional computation time becomes available.

The system also considers time-dependent transportation conditions and multiple optimization criteria. As a result, users can compare routes based on estimated travel time, fare, walking distance, transfers, and comfort.

6. CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The Intelligent Route Planner with Multi-Modal Transport aims to provide an efficient and user-friendly solution for finding suitable routes by integrating multiple transportation modes, including walking, cycling, buses, and metro services. The system combines graph-based pathfinding algorithms such as Dijkstra's Algorithm, A*, and Anytime Weighted A* to identify appropriate routes between a source and destination.

The proposed system considers several factors that influence travel decisions, including travel time, transportation cost, traffic conditions, waiting time, public transportation schedules, walking distance, and user comfort. By incorporating time-dependent costs and multi-objective optimization, the system enables users to compare different route alternatives and select the most suitable option based on their preferences.

The interactive map visualization provides a clear representation of routes, transportation segments, transfer points, and estimated journey details. This helps users make informed travel decisions while improving the flexibility of route planning.

Overall, the project demonstrates how artificial intelligence search techniques and graph algorithms can be applied to real-world transportation problems. The proposed approach provides a foundation for developing intelligent, adaptable, and personalized route-planning applications that support efficient urban mobility.

6.2 FUTURE WORK

The proposed system can be enhanced in several ways to improve its accuracy, usability, and practical application.

1. Real-Time Traffic Integration: Integrate live traffic data to update travel times dynamically and recommend alternative routes when congestion occurs.

2. Live Public Transportation Tracking: Incorporate real-time bus and metro arrival information to improve departure planning and reduce waiting time.

3. Machine Learning-Based Travel Prediction: Use machine learning techniques to predict travel duration, traffic congestion, and possible transportation delays using historical and current data.

4. Mobile Application Development: Extend the web application into an Android or iOS application with GPS-based navigation and location tracking.

5. Advanced Multi-Objective Optimization: Improve route selection by considering additional factors such as environmental impact, accessibility, safety, and passenger comfort.

6. Eco-Friendly Route Planning: Recommend routes that reduce carbon emissions by prioritizing walking, cycling, and suitable public transportation options.

7. Personalized Recommendations: Learn from user preferences to suggest routes that match individual requirements, such as minimum walking distance, lowest cost, or fewer transfers.

8. Voice-Based Navigation: Add voice input and turn-by-turn navigation to improve convenience while travelling.

9. Accessibility Support: Include wheelchair-accessible paths, elevators, ramps, and other facilities for passengers with special mobility requirements.

10. Performance Optimization: Evaluate and improve the performance of Dijkstra's Algorithm, A*, and Anytime Weighted A* on larger transportation networks using execution time, route quality, and memory consumption.

In the future, these enhancements can transform the proposed system into a comprehensive intelligent mobility platform that provides real-time, personalized, cost-effective, and environmentally sustainable transportation recommendations.

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